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# Streaming Multi-Agent Autoregressive Diffusion Model with World State Registers

*A Shared World State Register Enables Arbitrarily Long, Spatially Consistent Multi-Perspective Video Generation Without Quadratic Context Growth, Advancing Autonomous Driving Simulation*

**By** Sicheng Mo, Yuheng Li, Ziyang Leng, Krishna Kumar Singh, Bolei Zhou (University of California, Los Angeles)

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Autonomous driving simulators, multi-agent games, and embodied AI research all require generating photorealistic video from multiple agent perspectives that remain mutually consistent over long time horizons. Current video diffusion models generate single-agent views or suffer from consistency drift when extended across multiple agents and timesteps. A new paper from UCLA introduces a **Streaming Multi-Agent Autoregressive Diffusion (SMAD)** framework with **World State Registers (WSR)**—a fixed-size shared latent mechanism that aggregates and broadcasts global world state to all agents at each step, reducing Fréchet Video Distance by 29% and improving cross-agent consistency by 34% over strong baselines.

**AT A GLANCE**

**Problem:** Multi-agent video generation with independent per-agent diffusion models produces mutually contradictory world states that diverge over long rollouts.

**Why it matters:** Photorealistic, coherent multi-agent simulation is a critical infrastructure component for autonomous driving and embodied AI training at scale.

**Method:** Streaming autoregressive diffusion per agent, conditioned on a shared World State Register updated each step via a write-then-broadcast mechanism.

**Result:** FVD of 294 vs. 412 (independent baseline) and 0.82 vs. 0.61 cross-agent consistency score.

**Evidence:** Evaluated on a 6-camera surround autonomous driving benchmark across 100-step rollouts; ablations confirm WSR update frequency and register size are critical parameters.

## The Big Picture

Training autonomous vehicles, robots, and embodied agents at scale requires generating realistic sensor data—particularly video—from multiple co-located agents simultaneously. The gold standard is a world model that produces photorealistic, physically plausible multi-perspective video for any number of agents over arbitrarily long time horizons, with all agent views remaining mutually consistent.

Standard video diffusion models address single-agent scenarios. Naïve extension to multi-agent settings either (a) generates each agent's view independently—producing contradictory depictions of shared world elements—or (b) concatenates all agents' contexts as conditioning information, incurring memory costs that scale as $O(A \cdot T \cdot d)$ (agents $\times$ timesteps $\times$ dimension), making long rollouts intractable.

## Why Existing Approaches Fall Short

**Independent per-agent diffusion:** Without cross-agent coordination, two agents viewing the same intersection from different angles may generate inconsistent traffic light states, object positions, or pedestrian motions. These inconsistencies compound over time.

**Full history concatenation:** Conditioning each agent's diffusion model on all other agents' full history provides consistency at the cost of quadratic memory growth. For 6 cameras $\times$ 100 timesteps, this is computationally prohibitive.

**Fixed-frame world model:** Existing multi-camera driving world models generate fixed-length video clips rather than streaming generation, making them unsuitable for online simulation.

SMAD addresses all three limitations through the WSR mechanism.

## Core Mechanism

The **World State Register** is a compact set of $M$ latent vectors $\mathbf{s} \in \mathbb{R}^{M \times d}$ that encodes the current global state of the simulated world. At each generation step $t$:

1. Each of the $A$ agents generates its next frame conditioned on its local sliding window and the current register $\mathbf{s}_t$.
2. The register is **updated** via cross-attention over all agents' newly generated frame encodings.
3. The updated register $\mathbf{s}_{t+1}$ is broadcast to all agents for step $t + 1$.

Because the register has fixed size $M$ (independent of $A$ and $T$), per-agent memory cost is constant regardless of simulation length or the number of agents. The register serves as a sufficient statistic for cross-agent coordination:

each agent needs only the register, not other agents' raw frames.

## Technical Formulation

The joint generative distribution factorizes as:

$$p\left(x_{1:T}^{(1:A)}\right) = \prod_{t=1}^{T} p\left(\mathbf{s}_t \mid x_{<t}^{(1:A)}, \mathbf{s}_{t-1}\right) \prod_{a=1}^{A} p_\theta\left(x_t^{(a)} \mid x_{t-W:t-1}^{(a)}, \mathbf{s}_t\right)$$

where $W$ is the local history window size. Each per-agent model $p_\theta$ is a latent diffusion model with the register injected as cross-attention conditioning:

$$x_t^{(a)} \sim p_\theta\left(\cdot \mid x_{t-W:t-1}^{(a)}, \mathbf{s}_t, c^{(a)}\right)$$

where $c^{(a)}$ encodes agent-specific conditioning (camera pose, trajectory, goal). The register update is:

$$\mathbf{s}_{t+1} = \mathrm{Attn}\left(Q = \mathbf{s}_t,\ K = V = [\mathbf{s}_t;\ \phi_{\mathrm{enc}}(x_t^{(1)});\ \ldots;\ \phi_{\mathrm{enc}}(x_t^{(A)})]\right)$$

A register compression loss ensures the register faithfully captures world state:

$$\mathcal{L}_{\mathrm{reg}} = \mathbb{E}\left[\left\|\mathbf{s}_t - \phi_{\mathrm{enc}}\left(x_t^{(1:A)}\right)\right\|^2\right]$$

added to the diffusion denoising objective during training.

## Learning or Inference Procedure

Training uses paired multi-agent driving video from the nuScenes-derived MultiAgent benchmark (6 surround cameras, synchronized at 12 fps). The per-agent diffusion model and register update module are trained jointly end-to-end. The register is initialized to a learned null vector at the start of each sequence and is updated autoregressively without backpropagating through the full rollout history (stopped gradient on $\mathbf{s}_{t-1}$ during training).

At inference, generation proceeds streaming: no future frames are required, and the register can be updated in real time as frames are generated, enabling online simulation.

## What the Guarantee Says

The factorization structure guarantees that the register is the *only* information channel between agents at each step. This means that any information not captured in the $M$-vector register (e.g., fine-grained local details beyond what the update attention can aggregate) cannot be communicated across agents. Register size $M$ thus controls the fidelity–efficiency tradeoff.

## Experimental Findings

| Method | FVD ↓ | Consistency ↑ |
|---|---|---|
| Independent per-agent | 412 | 0.61 |
| Shared history concat. | 387 | 0.68 |
| **SMAD (WSR)** | **294** | **0.82** |

Long-horizon stability (normalized score at 100 steps) is 0.74 for SMAD vs. 0.43 for independent baselines, demonstrating that the WSR effectively prevents temporal drift over long rollouts.

## Ablations and Interpretation

**WSR update frequency:** Updating the register every step vs. every 4 steps degrades FVD from 294 to 361, confirming that frequent synchronization is necessary for maintaining consistency.

**Register size $M$:** FVD plateaus at $M = 64$ vectors. Below $M = 32$, cross-agent consistency collapses; above $M = 128$, there is no measurable benefit.

**Local window $W$:** Replacing the local $W = 8$ frame window with global history improves FVD marginally (288 vs. 294) but increases per-step memory by 12×, confirming the local window is an efficient approximation.

**Register initialization:** Random initialization produces $\Delta\mathrm{FVD} = +14$ vs. the learned null vector, confirming that a well-initialized register provides important context for the first few generation steps.

## Reference

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